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Pembuatan dan Mekatronik

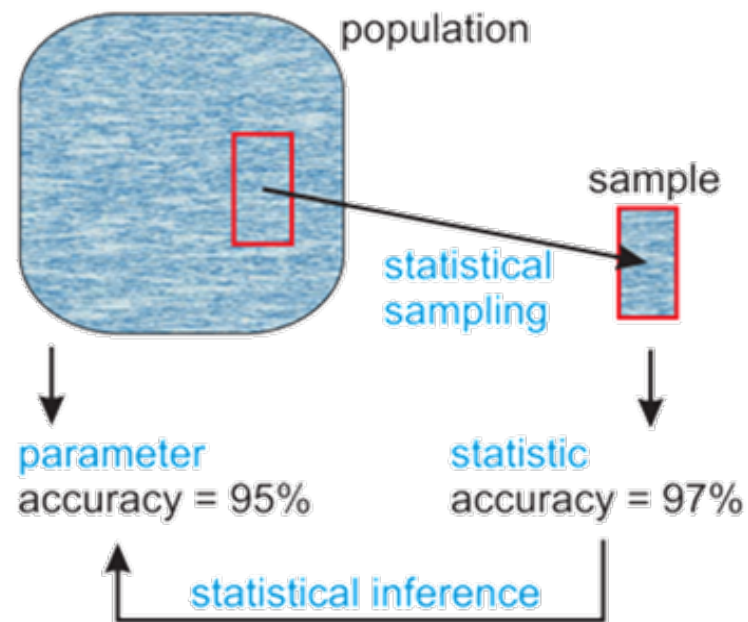
Fundamentals of Machine Machine Learning

**Beginner
Model Evaluation
(Slot 5)
25th May 2023**



Introduction

- Evaluation aims at selecting the most appropriate learning schema for a specific problem
- We evaluate its ability to generalize what it has been learned from the training set on the new unseen instances



Evaluation Strategies

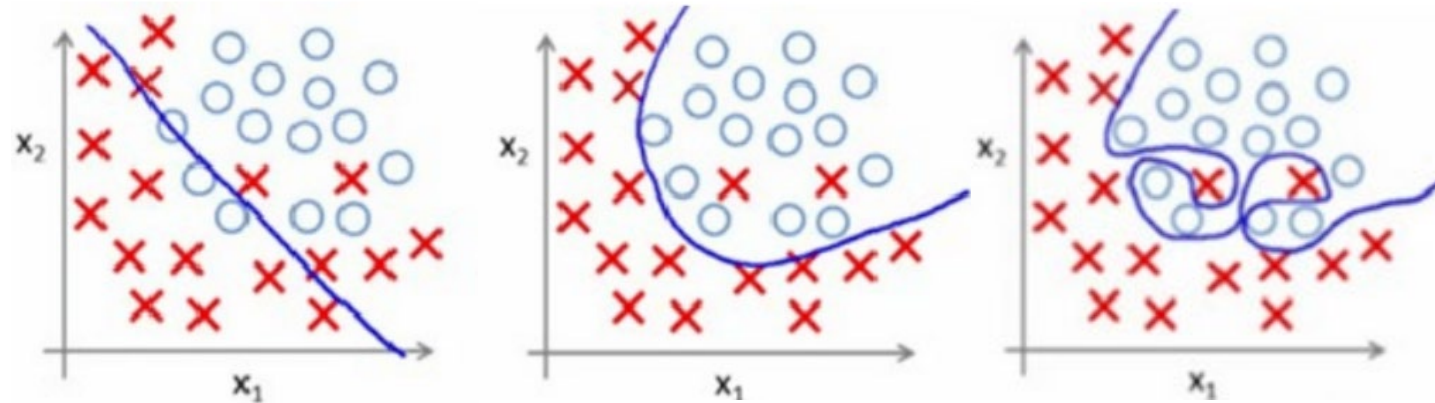
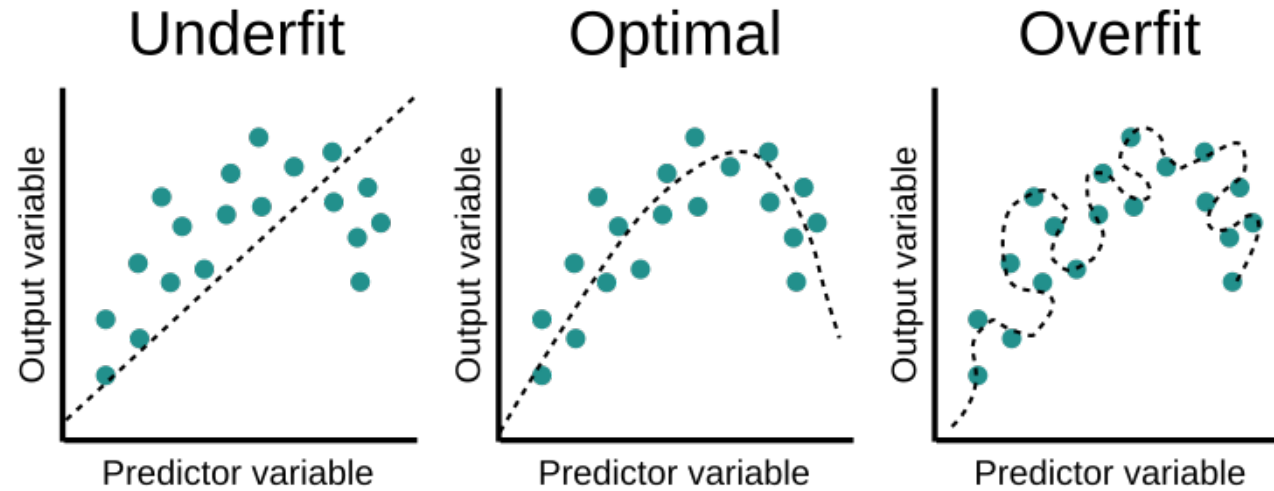
- Different strategies to split the dataset into training ,testing and validation sets.
- Error on the training data is not a good indicator of performance on future data, Why?
- Because new data will probably not be exactly the same as the training data!
- The classifier might be fitting the training data too precisely (called Over-fitting)

SELECTION OF BEST MODEL

- Generalisation
 - the ability of a machine learning model to perform well on new unseen data
- Overfitting
 - Overfitting refers to a model that models the training data too well.
 - Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.
 - This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.
- Underfitting
 - Underfitting refers to a model that can neither model the training data nor generalize to new data.
 - An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.

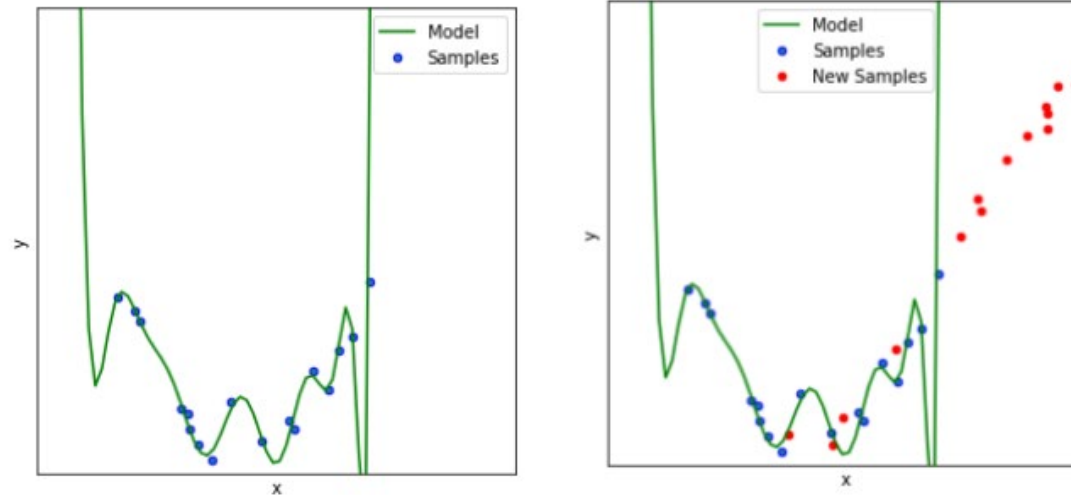
Evaluation Strategies

- Comparison on underfitting vs. optimal vs. overfitting



Check your understanding

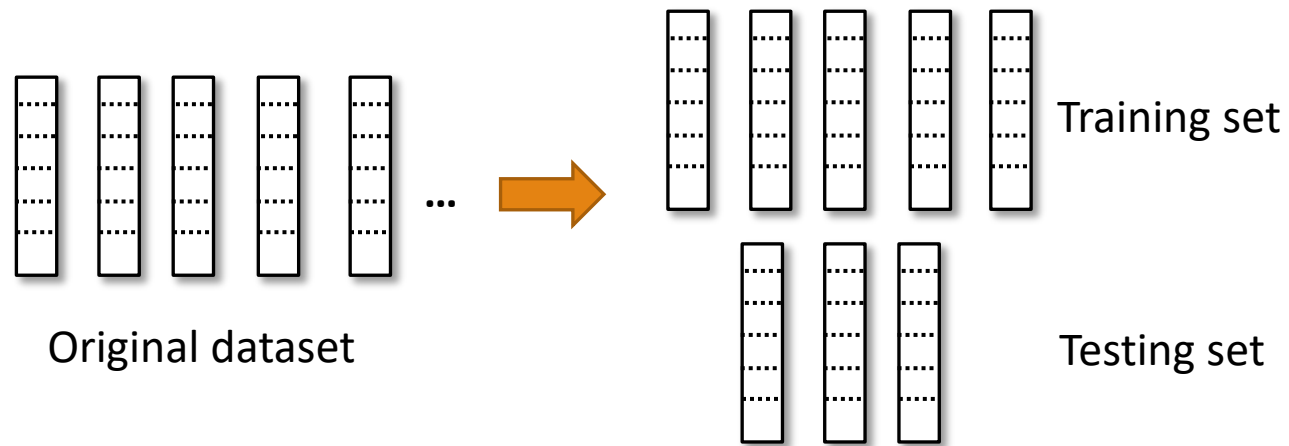
- In the graphs on the left below we can see a polynomial regression model curve (in green) which can predict almost every point (blue circles) on the training set exactly. But when new points (orange circles) are introduced in the graph to the right, it is unable to make accurate predictions on most of them.



- How you would you describe the model?
 - A. Underfitting
 - B. Overfitting
 - C. Neither, it generalizes perfectly well

Model evaluation methods

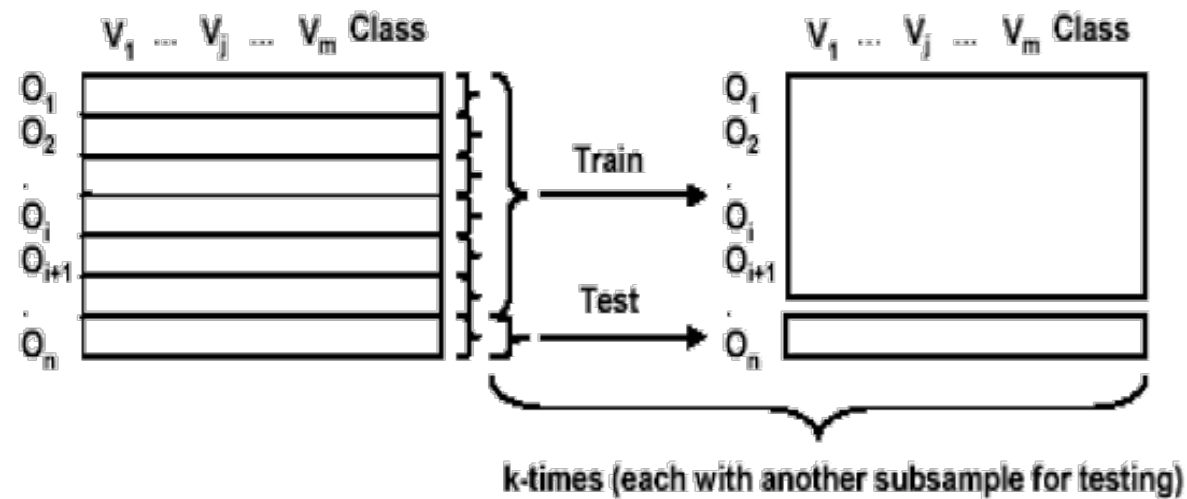
- Question: Since we hope to choose a model with smallest generalization error, how can we achieve our goal if we only have training samples?
- Answer: We can construct a testing set from the given dataset and use the testing error as the approximation of a model's generalization error.



- Methods to construct training set and testing set from the original dataset
 - Leave one out
 - Hold-out
 - Cross-validation
 - Bootstrapping

Model evaluation methods - Leave one out

- leave one instance out and train the data using all the other instances
- Repeat that for all instances
- compute the mean error



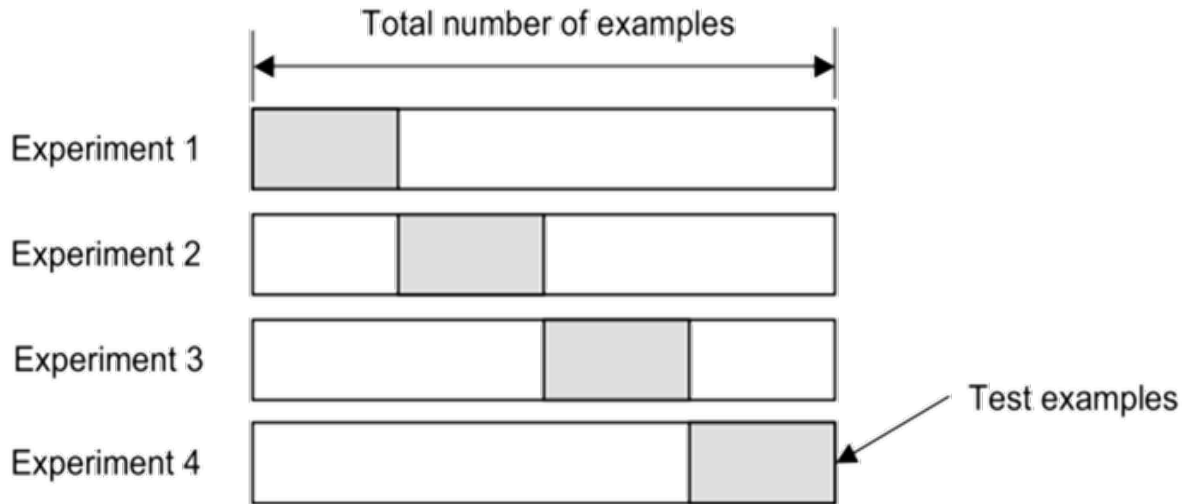
Model evaluation methods – Hold-out

- The dataset into two subsets use one of the training and the other for validation (In Large datasets)
- Common practice is to train the classifiers using 2 thirds of the dataset and test it using the unseen third.
- Repeated holdout method: process repeated several times with different subsamples
- error rates on the different iterations are averaged to give an overall error rate

Model evaluation methods – K-Fold Cross Validation

- Divide the data to k equal parts (folds),
- One fold is used for testing while the others are used in training,
- The process is repeated for all the folds and the accuracy is averaged.

- 10 folds are most commonly used
- Even better, repeated
- Cross-validation



Model evaluation methods – Bootstrap sampling

- CV uses sampling without replacement
- The same instance, once selected, can not be selected again for a particular training/test set
- The bootstrap uses sampling with replacement to form the training set
 - Sample a dataset of n instances n times with replacement to form a new dataset of n instances
 - Use this data as the training set
 - Use the instances from the original dataset that don't occur in the new training set for testing

Model Performance Evaluation

- Confusion Matrix

		Predicted	
		Yes	No
Actual	Yes	True Positive	False Negative
	No	False Positive	True Negative

- Performance evaluation:

- 1) Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- 2) Precision

$$Precision = \frac{TP}{TP + FP}$$

- 3) Recall

$$Recall = \frac{TP}{TP + FN}$$

- 4) F1-Test

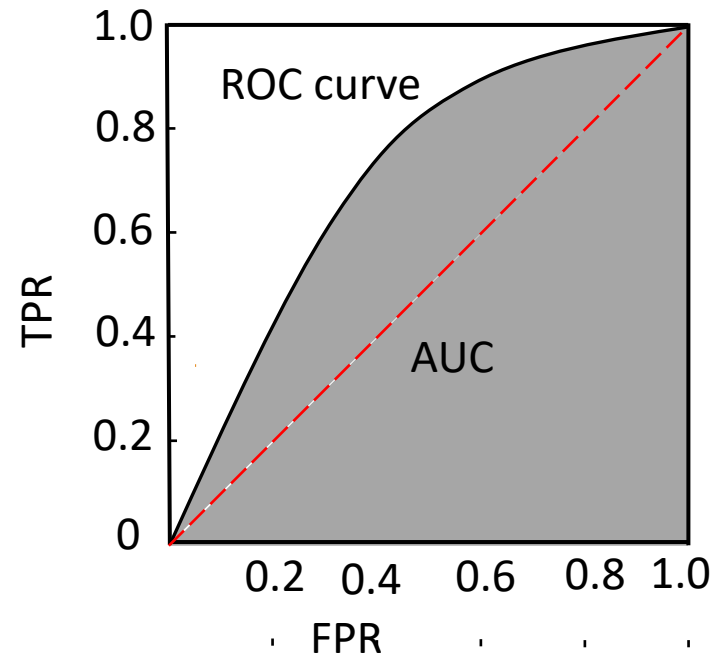
$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Model Performance Evaluation – ROC Curve

- Stands for “receiver operating characteristic”
- Used in signal detection to show tradeoff between hit rate and false alarm rate over noisy channel

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{TN + FP}$$



Hyperparameter Tuning

- Hyperparameters = are all the parameters which can be arbitrarily set by the user before starting training (eg. number of estimators in Random Forest).
- How to find good hyperparameter?
 - Manual search (trial and error)
 - Grid search
 - Random search

Hyperparameter Tuning – Manual search

- When using Manual Search, we choose some model hyperparameters based on our judgment/experience.
- We then train the model, evaluate its accuracy and start the process again.
- This loop is repeated until a satisfactory accuracy is scored.

Hyperparameter Tuning – Grid search

- In Grid Search, we set up a grid of hyperparameters and train/test our model on each of the possible combinations.
- In order to choose the parameters to use in Grid Search, we can now look at which parameters worked best with Random Search and form a grid based on them to see if we can find a better combination.
- When using Grid Search, all the possible combinations of the parameters in the grid are tried.

Hyperparameter Tuning – Random search

- In Random Search, we create a grid of hyperparameters and train/test our model on just some random combination of these hyperparameters.
- In order to choose the parameters to use in Grid Search, we can now look at which parameters worked best with Random Search and form a grid based on them to see if we can find a better combination.
- When using Grid Search, all the possible combinations of the parameters in the grid are tried.